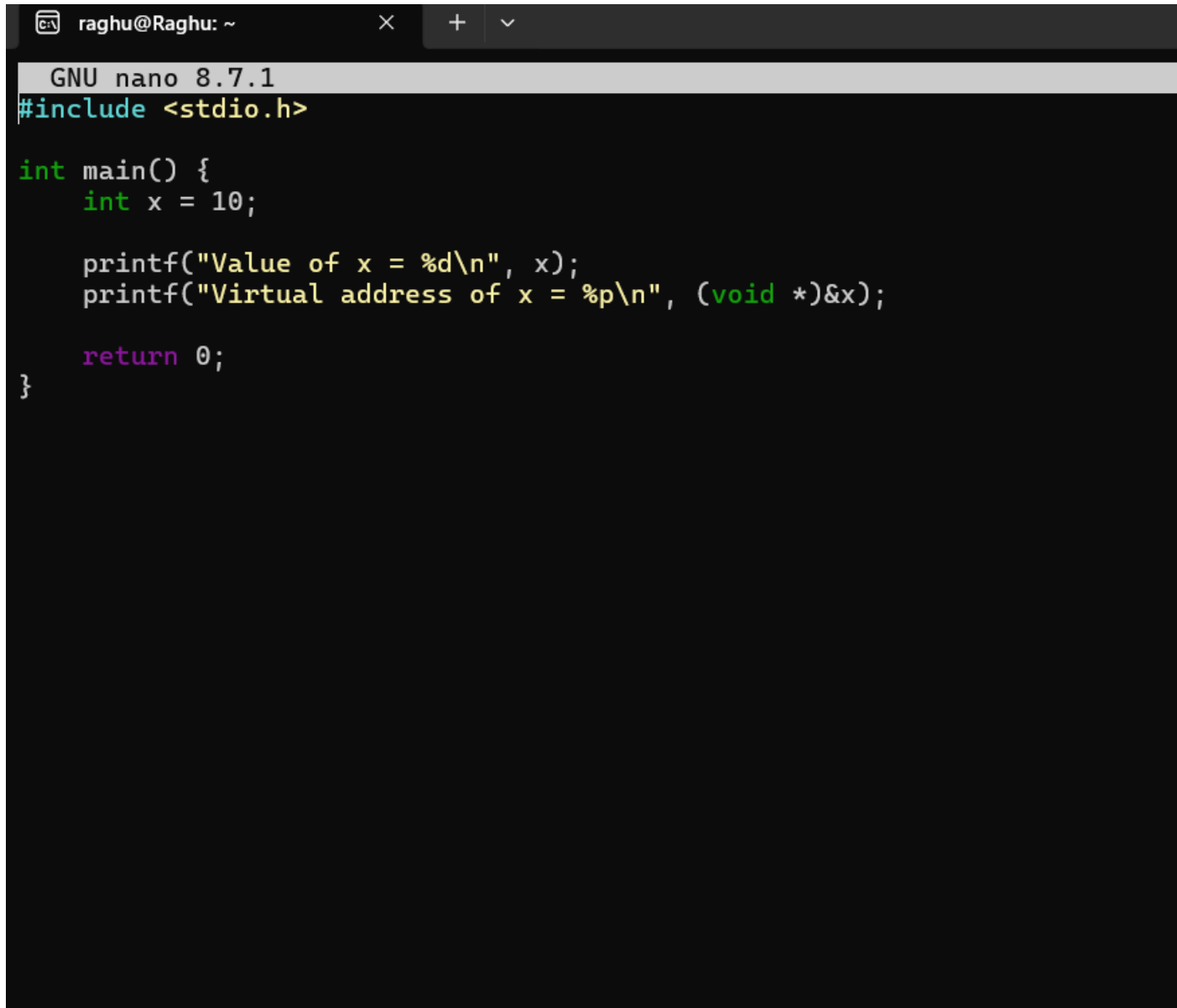


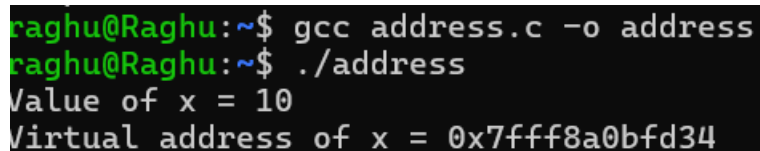
2520030225: R.RAGHUVVEER

1. address.c



```
raghu@Raghu: ~  
GNU nano 8.7.1  
#include <stdio.h>  
  
int main() {  
    int x = 10;  
  
    printf("Value of x = %d\n", x);  
    printf("Virtual address of x = %p\n", (void *)&x);  
  
    return 0;  
}
```

Out put:



```
raghu@Raghu:~$ gcc address.c -o address  
raghu@Raghu:~$ ./address  
Value of x = 10  
Virtual address of x = 0x7fff8a0bfd34
```

2. memory_segments.c

```
raghu@Raghu: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <stdlib.h>  
  
int global = 10;  
int bss_variable;  
  
int main() {  
  
    int local = 20;  
  
    int *heap = malloc(sizeof(int));  
    *heap = 30;  
  
    printf("Code address   : %p\n", (void *)main);  
    printf("Data address   : %p\n", (void *)&global);  
    printf("BSS address    : %p\n", (void *)&bss_variable);  
    printf("Heap address   : %p\n", (void *)heap);  
    printf("Stack address  : %p\n", (void *)&local);  
  
    free(heap);  
  
    return 0;  
}
```

Output :

```
raghu@Raghu:~$ nano memory_segments.c  
raghu@Raghu:~$ gcc memory_segments.c -o memory_segments  
raghu@Raghu:~$ ./memory_segments.c  
-bash: ./memory_segments.c: Permission denied
```

3:

```
raghu@Raghu: ~  
GNU nano 8.7.1  
#include <stdio.h>  
  
int main() {  
    int page_table[4] = {1, -1, 3, -1};  
    int page;  
  
    printf("Enter page number (0-3): ");  
    scanf("%d", &page);  
  
    if (page_table[page] == -1) {  
        printf("Page Fault!\n");  
        printf("Page must be loaded into memory.\n");  
    }  
    else {  
        printf("Page is already in memory.\n");  
        printf("Physical frame = %d\n", page_table[page]);  
    }  
  
    return 0;  
}
```

Output:

```
raghu@Raghu:~$ nano page_fault.c  
raghu@Raghu:~$ gcc page_fault.c -o page_fault  
raghu@Raghu:~$ ./page_fault  
Enter page number (0-3): 1\  
Page Fault!  
Page must be loaded into memory.
```

4:

```

raghu@Raghu: ~
GNU nano 8.7.1
#include <stdio.h>

int main() {

    int page_table[] = {3, 5, 1, 7};
    int page_size = 100;

    int virtual_address = 250;

    int page = virtual_address / page_size;
    int offset = virtual_address % page_size;

    int frame = page_table[page];

    int physical_address = frame * page_size + offset;

    printf("Virtual Address = %d\n", virtual_address);
    printf("Page Number      = %d\n", page);
    printf("Offset           = %d\n", offset);
    printf("Frame Number      = %d\n", frame);
    printf("Physical Address = %d\n", physical_address);

    return 0;
}

```

Output :

```

raghu@Raghu:~$ nano address_translation.c
raghu@Raghu:~$ gcc address_translation.c -o address_translation
raghu@Raghu:~$ ./address_translation.c
-bash: ./address_translation.c: Permission denied
raghu@Raghu:~$ address_translation
address_translation: command not found
raghu@Raghu:~$ gcc address_translation.c -o address_translation
raghu@Raghu:~$ ./address_translation
Virtual Address = 250
Page Number      = 2
Offset           = 50
Frame Number      = 1
Physical Address = 150
raghu@Raghu:~$ |

```

5:

```
raghu@Raghu: ~  
GNU nano 8.7.1  
#include <stdio.h>  
#include <stdlib.h>  
  
int main() {  
  
    int *p;  
  
    p = malloc(5 * sizeof(int));  
  
    if (p == NULL) {  
        printf("Memory allocation failed\n");  
        return 1;  
    }  
  
    for (int i = 0; i < 5; i++) {  
        p[i] = (i + 1) * 10;  
    }  
  
    for (int i = 0; i < 5; i++) {  
        printf("%d ", p[i]);  
    }  
  
    printf("\n");  
  
    free(p);  
  
    return 0;  
}
```

Output :

```
raghu@Raghu:~$ nano malloc_array.c  
raghu@Raghu:~$ gcc malloc_array.c -o malloc_array  
raghu@Raghu:~$ ./malloc_array  
10 20 30 40 50  
raghu@Raghu:~$
```

6:

```
raghu@Raghu: ~  
GNU nano 8.7.1  
#include <stdio.h>  
  
int main() {  
    int *p = NULL;  
    printf("Before accessing memory\n");  
    *p = 100;  
    printf("After accessing memory\n");  
    return 0;  
}
```

Output:

```
raghu@Raghu:~$ nano malloc_array.c  
raghu@Raghu:~$ nano null_pointer.c  
raghu@Raghu:~$ gcc null_pointer.c -o null_pointer  
raghu@Raghu:~$ ./null_pointer  
Before accessing memory  
Segmentation fault ./null_pointer
```